

Next week, **\$604,163** of rake walks out — and we have the names.

An online poker room does not gamble. It takes a small cut of each pot — the **rake** — and earns nothing at all unless recreational players keep sitting down. We rebuilt **21,605,687 real hands** into a player-level ledger, predicted who goes quiet next week, priced what each of them is worth to the business, and multiplied the two.

RAKE AT RISK NEXT WEEK

\$604,163

58,836 of 77,268 players are more likely than not to play zero hands next week

REACHABLE AT A 10% CONTACT BUDGET

\$421,064

7,727 calls, ranked by expected loss — 70% of the money at risk

NAMES IT SHARES WITH A CHURN-RISK LIST

0.1%

Sorting the same players by churn risk produces a different list that reaches **0.8%** of the money

Weekly rake in view **\$2,114,626**, of which **34.4%** is measured against the pot and the rest priced from what comparable hands actually paid · six commercial venues, iPoker excluded from all money questions (its rake reconciles on none of its 6.0 M hands) · July 2009, **2009 US dollars**, no currency conversion and no inflation adjustment · “leaving” means **zero hands in the next seven days**, measured against each venue's own last recorded day.

THE BUSINESS PROBLEM

The house never wins a hand. It *rents the seat* — and the tenants are leaving.

A poker operator has no stake in who wins. It takes roughly a twentieth of each pot — measured across all six venues in this data, the median rake is **4.3%–5.0% of the pot**, and **exactly \$0.00 when no flop is dealt**. Revenue is therefore a pure function of how many hands get played, which is a pure function of how many losing players are willing to keep sitting down.

01 · SUPPLY

Recreational players deposit

They play loose, lose steadily, and treat the loss as the price of the evening.

–14.3 bb/100

02 · EXTRACTION

Professionals take it

A small group plays a dozen tables at once and wins nearly everything the recreational players lose.

11.5 tables each

03 · THE BUSINESS

The operator rakes every pot

It earns from the churn of money between the two groups — not from either of them winning.

\$2,114,626 a week

04 · COLLAPSE

The supply stops

When recreational players quietly stop logging in, the professionals have nobody to beat — and the rake dies with the games.

no players, no pots, no rake

THE QUESTION THIS PROJECT ANSWERS

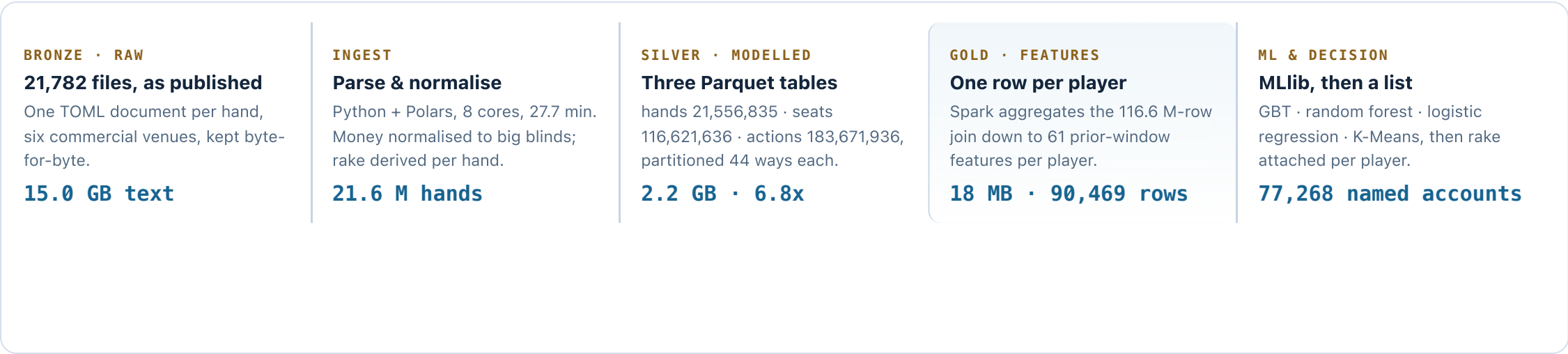
Which players will stop playing next week — and how much of next week's rake leaves with them?

Both halves matter. A churn score alone cannot tell an operator whether the player about to leave was worth \$13 a week or \$221. The deliverable is one ranked, named, exportable list — not a metric.

WHY THIS IS A RETENTION PROBLEM, NOT A POKER PROBLEM

Nothing here models how to win at poker, and nothing identifies a player to exploit. The same list that protects revenue is the list a responsible-gaming team would want: it flags the customers who are losing fastest and disengaging quietest. **We rank who to contact and protect, never who to target.**

116,619,267 seat-rows in. *18 MB out.* The models never touch the big data.



VERIFIED — PRINTED BY THE BUILD, NOT CLAIMED

21,556,435 distinct hands + **49,252** dropped with a recorded reason = **21,605,687** — the dataset's published count, exactly. Total loss **0.23%**, every hand accounted for.

THE TEST THAT CAUGHT A REAL BUG

Poker is zero-sum apart from the rake, so $\sum \text{net_bb}$ must equal $-\sum \text{rake_bb}$. Measured at full scale: **-2,064,330 vs -2,064,329**, ratio 1.0000. Before the uncalled-bet fix this identity failed on 99.4% of hands.

Local Spark (local[8], 9 GB driver) on one laptop; the logic re-hosts to Databricks unchanged.

WHY THE COMPRESSION IS THE ARGUMENT

Excel stops at 1,048,576 rows. The seat table is **111x** that and the action table **175x**. But the table the models read is **18 MB** — the distributed work happens once, in feature engineering, and never again.

SCHEMA-ON-READ, HONESTLY

The same field arrives as `int` in one hand and `float` in the next, and one venue writes string hand IDs. Types are declared explicitly rather than inferred — inference killed two full runs before that changed.

WHAT WE REFUSED TO IMPUTE

One venue writes the vendor's name into every table field, so its multi-tabling count is a missing value wearing a plausible number. It is **nulled, not filled with 1** — filling it would file 13,201 professionals as single-table recreational players.

We lose the players we earn from, and keep *the ones we don't*.

Four clusters over volume, multi-tabling and betting style, on **77,268** players. Nothing about leaving was in the feature set; the lapse column was cross-tabbed afterwards.

SEGMENT	SHARE OF PLAYERS	HANDS	TABLES	BB/100	LAPSE RATE	RAKE AT RISK
Recreational	41.2% 31,799	352	1.4	-14.3	77.1%	\$182,482 30.2% · \$13.59/head
Gambler	19.8% 15,262	331	1.3	-31.8	74.6%	\$157,897 26.1% · \$24.91/head
Regular	35.4% 27,376	884	3.0	-7.2	61.6%	\$155,173 25.7% · \$24.73/head
Grinder	3.7% 2,831	8,799	11.5	-3.3	61.3%	\$108,610 18.0% · \$220.88/head

THE TWO THAT PAY, LEAVE

Recreational and Gambler are **60.9%** of players but **65.9%** of everyone who goes quiet — and they hold **56%** of the rake at risk (\$340,379).

THE ONE THAT DOESN'T, STAYS

The Grinder loses least — **-3.3** bb/100, essentially just the rake — and is the stickiest player on the platform. Per head a Grinder pays **\$220.88** of weekly rake against a Recreational player's **\$13.59**. That measures who *paid* it, not whose money funded it.

WHY MULTI-TABLING EARNS ITS PLACE

On betting style alone **no professional cluster forms at all** — two of four clusters both land on “recreational”. Style cannot separate a professional from a customer; **table count can**. That single feature is why the segmentation is usable.

WHAT WE WILL NOT OVERSELL

These are **gradients, not species**. Silhouette actually peaks at k=2 (0.439); k=4 costs about 0.12 and was chosen for business readability. **Bisecting k-means scores better** (0.319 vs 0.228) and the two algorithms place only **67%** of players in the same group. Two of our four hypothesised archetypes were also wrong: no ultra-tight “nit” cluster exists, and the loose cluster is **passive, not aggressive**. We report the misses.

Going quiet is *predictable a week ahead* — from behaviour recorded before the week starts.

Label: **zero hands in the next seven days**, measured against each venue's own last recorded day, because the six venues do not share a calendar. Every feature is recomputed from hands on or before the cutoff, so the model structurally cannot see the week it is predicting.

THE FINDING THAT CHANGED THE POPULATION

The obvious filter — “players with at least 100 hands” — counts hands over a player's **whole history, including the week being predicted**. Someone with 30 prior hands only clears it by playing 70 more *in the label week*, which means by definitely not lapsing.

The arithmetic is decisive: the lifetime rule keeps **98,718** players, the prior-window rule keeps **90,469**, and **both contain exactly 61,702 lapsers**. Every one of the 8,249 extra players is a non-lapser admitted by a filter that had already seen the answer. Prevalence moves from 62.5% to **68.2%** once it is fixed.

THE LEAKAGE CHECK, RE-RUN ON THE HONEST FEATURES

Same estimator, same players, same split. Volume and tenure alone: **0.690**. Lifetime rates, which can see the label week: **0.819**. Prior-window only: **0.857**. **The honest feature set wins** — the earlier ordering had been the other way round, which is the signature of leakage.

WHAT THE MODEL IS GIVEN · 30 PRIOR-WINDOW FEATURES

- **Recency & rhythm** — days since last hand, share of tenure days actually played, minutes at the table, hands per minute.
- **Decay** — hands in the last week against the week before, and the share of all volume falling in the final three days.
- **Style** — VPIP, pre-flop raise, aggression factor, showdown rate, average investment per hand.
- **Context** — stakes played and their spread, seats per table, button share, and a dummy per venue.

Recency is the strongest single feature (GBT importance 0.249) and the next is worth a third as much. We say so rather than overselling the behavioural columns.

LOGISTIC

0.837

readable coefficients

RANDOM FOREST

0.857

a statistical tie with GBT

GBT · SHIPPED

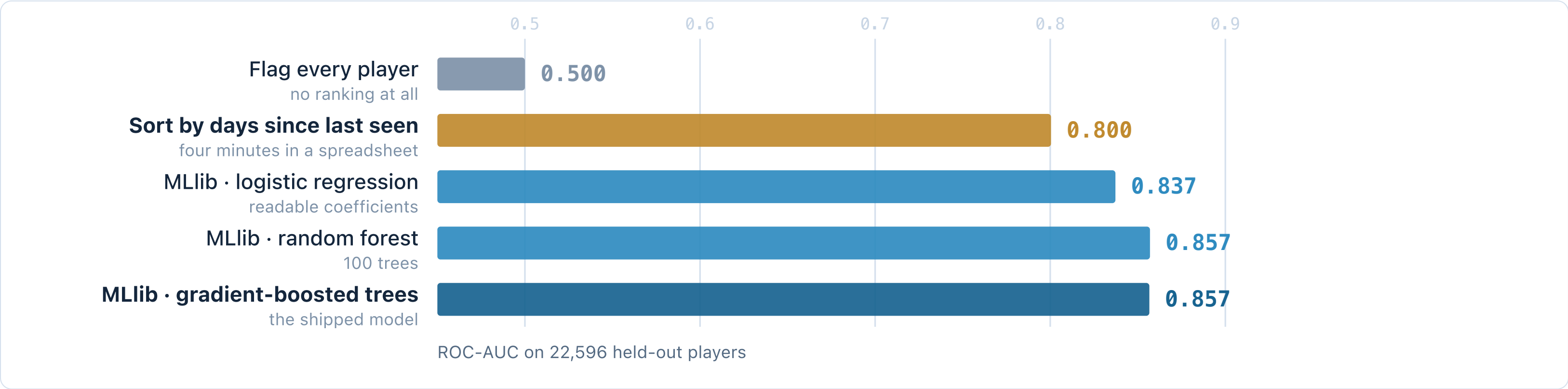
0.857

ROC-AUC, 22,596 held out

WHERE IT NEARLY FAILED

A four-minute spreadsheet sort scores **0.800**. At its default setting, *our model loses to it*.

Sorting players by days-since-last-seen needs no model and no training. It is the bar every line of this pipeline has to clear, so it is on the slide rather than in an appendix.



THE HONEST LOSS

At the default 0.5 cut-point, the rule “quiet for a day” scores **0.856** on F1 and our GBT scores **0.829**. On a 68/32 problem, 0.5 is an inherited default, not a decision. Chosen deliberately, GBT reaches **0.867** at a threshold of **0.40**.

WHERE THE MODEL DOES EARN ITS PLACE

Under a real contact budget. At **2,260** calls the model finds **2,192** true lapsers against the free sort's **2,001** — **191** fewer offers posted to people who were never leaving, every week.

WHAT WE ARE NOT CLAIMING

The split is by player, never by period, so **no temporal backtest exists** and we do not claim the model generalises to a future month. It is also weakest where lapsing is rarest (0.769 on Ogame against 0.861 on PokerStars). Both are named limitations, not discovered ones.

Twenty-six days of data cannot show that anyone left for good. Everything here is **short-horizon lapse — a week of silence, an early warning** — and it is called that throughout.

The model ranks players. It does not price them — and *both halves of the multiplication were broken.*

DEFECT 1 · RANKED RIGHT, PRICED WRONG

The classifier was fitted with balanced class weights, which drags every predicted probability toward 0.5. ROC-AUC cannot see it — it only reads order — but a dollar figure multiplies by that number directly.



At 68/32 there was never an imbalance to correct. Removing the weights moved the worst decile from **0.200** off to **0.026** while ROC-AUC did not move (0.857 to 0.857). A 0.03 tolerance was written into the code before the run, so no calibration layer was needed.

DEFECT 2 · RAKE IS DERIVED, AND THE HANDS THAT RECONCILE ARE THE BIG ONES

Rake is not a recorded column anywhere in this data. It is pot – paid-out winnings, and on venues whose exports are incomplete that residual only looks sane on large pots. On PartyPoker the hands where it reconciles have a **median pot of 23.4 bb and 99.5% saw a flop**; the rest, **2.5 bb and 39.0%**.

So the obvious method — scale each player's observed rake per 100 hands to all their hands — charges every hand at the rate of the biggest pots. Measured, it runs **25%** high.

THE SCREEN, AND THE ESTIMATOR THAT REPLACED THE SHORTCUT

PokerStars reconciles completely and **not one of its 1,107,219 flop hands rakes above 5.0% of the pot**, so anything above 6% is missing money wearing a rake's clothes. Everything else is priced by conditional mean **in big blinds** over (venue, stake, flop, pot size), converted to dollars only at the end. Fitted on half the measured hands and scored against the other half, it lands within **0.7% on every venue**.

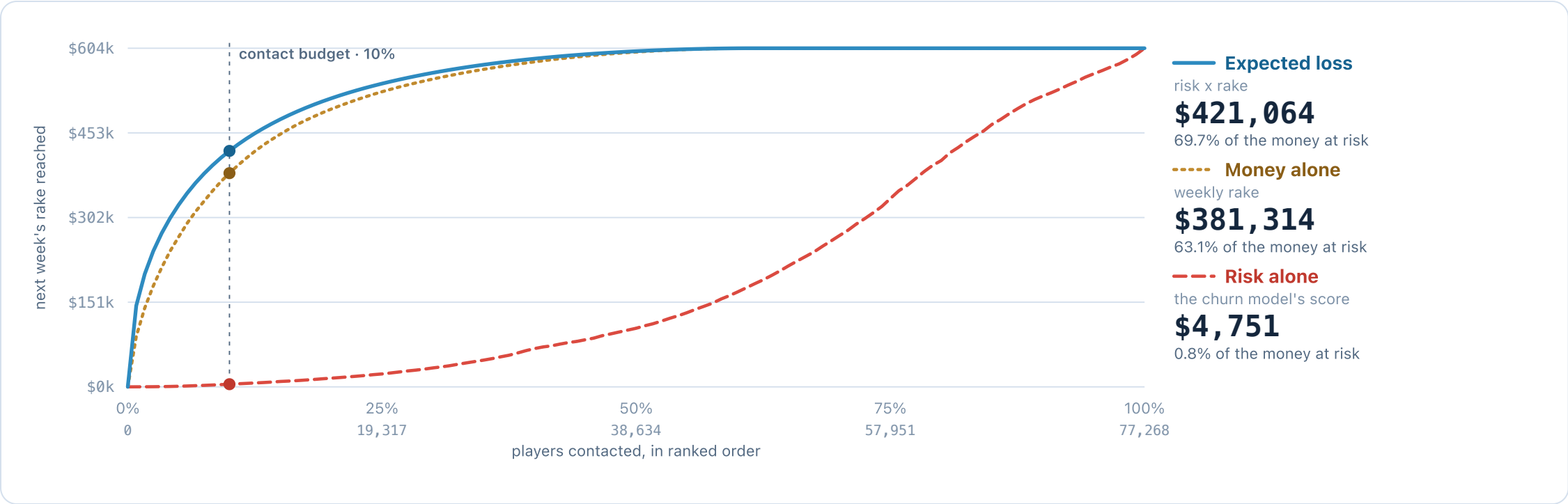
MEASURED
34.4%
of weekly rake reconciles against the pot

UNCALIBRATED WE WOULD HAVE SAID
\$424,810
understating the business case by **29.7%**

ATTRIBUTION CHECK
1.000000
contributed vs equal-split totals, \$8,217,411 both ways

THE RECOMMENDATION

A retention list sorted by churn risk is *almost exactly the wrong list*.



THE LIST THAT IS BEST AT ITS JOB AND WORST AT OURS

Ranking by risk alone finds **7,554** genuine leavers out of 7,727 calls — **98%** precision, far better than the list we recommend. It reaches **0.8%** of the money. It is a list of certain departures worth nothing.

WHY THE PRODUCT BEATS EITHER FACTOR

Money alone reaches **63.1%** but wastes two thirds of its calls on players who were not going anywhere. **Risk x value reaches 70%**, and the two lists share **0.1%** of their names.

CHECKED ON PLAYERS THE MODEL NEVER SAW

Repeated on the held-out fold alone — 19,360 players, 1,936 calls, labels no model ever trained on — the picture is identical: **70.0% of the money against 0.9%**, on **0.2% shared names**. The recommendation does not depend on players the model had already met.

SORT THE LIST BY	RAKE REACHED	SHARE OF THE MONEY	TRUE LEAVERS FOUND	PRECISION	NAMES IN COMMON
Expected loss risk x weekly rake	\$421,064	69.7%	4,146	54%	100.0%
Money alone rank by weekly rake	\$381,314	63.1%	2,521	33%	67.4%
Risk alone rank by the churn score	\$4,751	0.8%	7,554	98%	0.1%

Same 7,727 calls (10% of 77,268 players), three ways of choosing who gets them.

Curves and table recomputed from the shipped Gold table by `src/deck_charts.py`, which checks itself against the model run's own JSON before writing. One figure moves by design: `risk_cal` ships rounded to four decimals, so ties in the risk-only ranking break differently here than during the run — \$4,751 against \$4,729, 0.8% of the money either way.

Four decisions, each with *a number* and an owner.

01 Rank Monday's call list by expected loss, not by churn risk.

The same **7,727** calls reach **\$421,064** of next week's rake instead of **\$4,751**.

OWNER · RETENTION & CRM LEAD

02 Spend the retention budget on the recreational seat, not the top spender.

Recreational and Gambler hold **56%** of the rake at risk (**\$340,379**) against the professionals' 18% — and they are the money the professionals are paid from.

OWNER · VIP & PROMOTIONS

03 Run the list per venue. Never pooled.

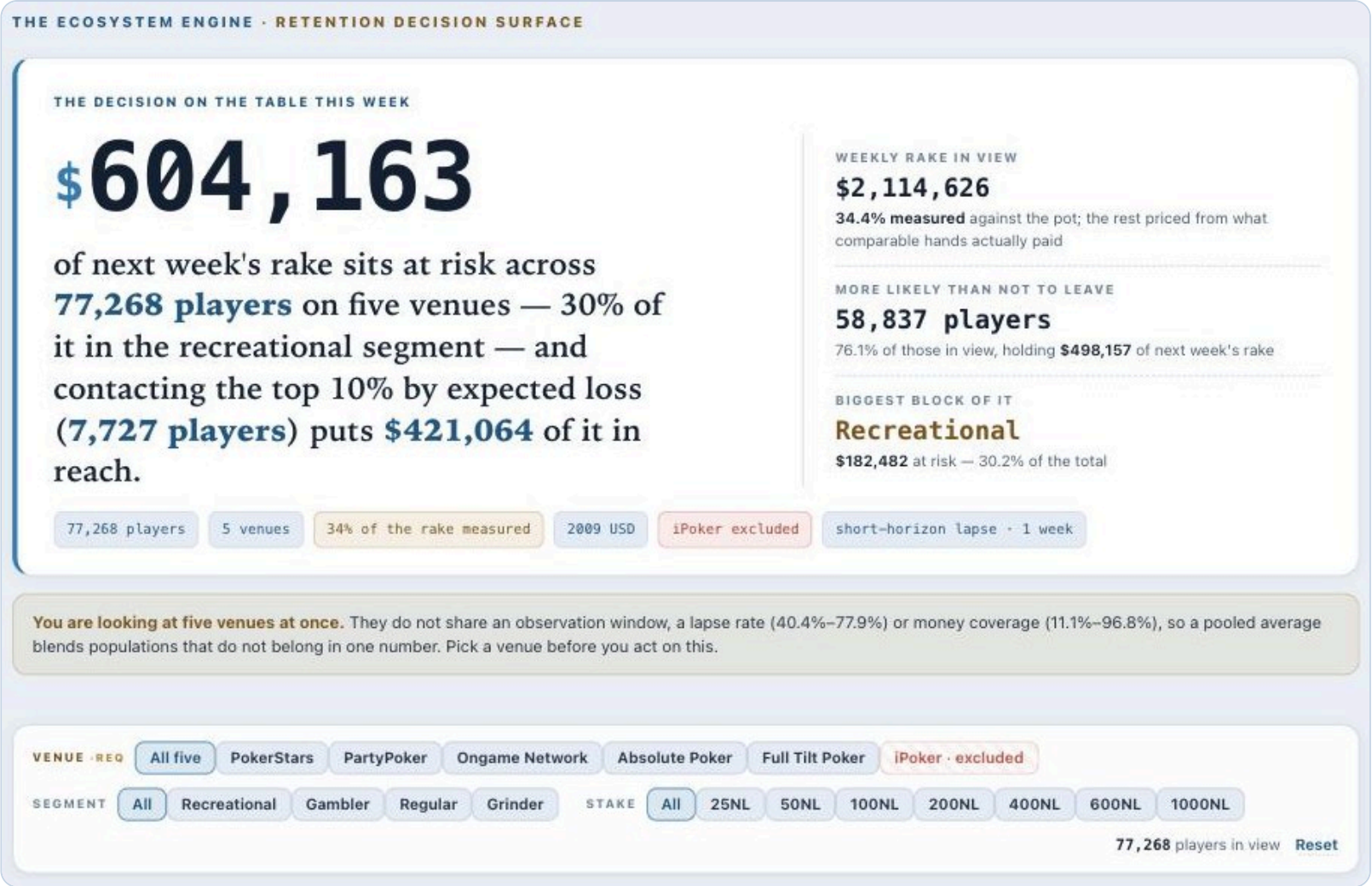
Lapse rates run **40.4% to 77.9%** and money coverage **11% to 97%** across the five venues. One venue is **45%** of the exposure; a pooled average describes none of them.

OWNER · REGIONAL OPERATIONS

04 Record rake at source, and this stops being an estimate.

Two thirds of the weekly rake is modelled because no venue writes it down. One logged column per hand would retire the largest caveat in this deck.

OWNER · DATA ENGINEERING



THANK YOU

One number, and the five things we will *not* hide behind it.

THE NUMBER

\$604,163

of next week's rake sits with players who are more likely than not to stop playing — 30% of it in the recreational segment, and \$421,064 of it reachable with 7,727 phone calls.

THE CAVEATS TRAVEL WITH IT

- July 2009. Seventeen years old. The method transfers; the magnitude is a micro-stakes market of its time.
- Rake is derived, never recorded — 34.4% measured, the rest priced from comparable hands.
- Twenty-six days. This is short-horizon lapse, not churn. Nobody in this data can be shown to have left for good.
- No ground truth for bots — which is exactly why the supervised question is lapse, and why bot detection is not claimed.
- Player IDs are per venue. The same human on two sites is two people here, and we never merge them.

IN THE REPOSITORY

src/ — ten scripts, Bronze to the call list, each re-runnable

WRITTEN UP

docs/ — plan, pipeline guide, and one results note per model with its own weaknesses

THE TOOL

docs/dashboard.html — self-contained, opens from a file, no server

REPRODUCIBILITY

seed 42 · every figure here injected from the run's own JSON by src/deck_charts.py

Questions — we would rather be asked about the caveats than the number.

TEAM: add names & roll numbers